

Gallium-related defects in *p*-type silicon irradiated with electrons and protons. Anisotropy of conductivity due to boron- and gallium-related defects in irradiated silicon

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ABSTRACT

Hall effect and conductivity measurements over a wide temperature range of 20–300 K are carried out on gallium-doped silicon subjected to 3.5 MeV electron and 15 MeV proton irradiation at room temperature. Electrical data obtained provide convincing evidence that impurity atoms are involved in interactions with intrinsic point defects during irradiation. As a result, heavy losses of the shallow acceptors in the irradiated samples are found. Then, to shed light on the electrical properties of the radiation-produced defects, these samples are subjected to annealing. Two kinds of defects, Ga–vacancy pairs and substitutional Ga[−]–interstitial Ga⁺ ion pairs, are earlier discussed in the literature. However, a new kind of Ga-related radiation defects that are electrically neutral in *p*-type materials makes its appearance as well. In fact, under our irradiation conditions, they turn out to be dominant and stable up to 450 °C. Interstitial Ga-related defects in the irradiated samples are believed to be responsible for the anisotropic effects observed for the hole mobility at cryogenic temperatures. Similar anisotropic effects are also observed in boron-doped silicon subjected to electron and proton irradiation. Some interesting peculiarities in the formation processes of interstitial Ga-related defects are discussed.

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I. INTRODUCTION

Among all shallow acceptor impurities in silicon, a large body of research information about impurity-related point defects has been gained for boron as the principal dopant in this semiconductor for wide use in the electronic industry. In contrast, our understanding of gallium impurity atoms and their complexes in silicon is more challenging, although the detection of several gallium-related defects in early radiation experiments seemed to be promising. Actually, ion pairs of substitutional and interstitial impurity atoms $[Ga_s^- Ga_i^{2+}]$ were first observed in the spectra of Electron Paramagnetic Resonance (EPR) taken in Ga-doped Si samples irradiated with 1.0 MeV electrons and then annealed at $T \approx 180$ °C.¹

Interestingly, there were no EPR signals just after irradiation and before annealing. Based on this observation, the indirect conclusion made was that the migration of interstitial Ga ions resulting in the formation of ion pairs $[Ga_s^- Ga_i^{2+}]$ starts at around $T \approx 200$ °C, whereas at lower temperatures, these ions were in a diamagnetic charge state, and hence, they were not observable by EPR.

Somewhat later, a comparative annealing study of radiation defects was carried out on boron-, aluminum-, and gallium-doped Si irradiated with 1.5 MeV electrons at $T = 78$ and 300 K.² In particular, an activation energy of $E_{act} = (0.66 \pm 0.05)$ eV for annealing stage 3 over a temperature range of $460 \leq T \leq 560$ K was determined in irradiated *p*-Si doped with Ga. This stage was tentatively

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identified as belonging to the annealing of “substitutional Ga atom–vacancy” complexes, $[Ga_s V]$. Of interest is the final high-temperature annealing stage of $T \approx 575$ K, which led reportedly to the complete restoration of lightly irradiated samples. Many modeling concepts of radiation defects in Si have been changed since then, so any tentative defect identification must be accepted using great caution. A long way of the boron-related radiation defect identification provides a typical example (see, for instance, Refs. 3 and 4).

Later, the use of Deep Level Transient Spectroscopy (DLTS) on Ga-doped floating zone Si samples [from here on p -Si(FZ):Ga] has found new dominant traps of majority charge carriers at $\approx E_V + 0.27$ eV and $\approx E_V + 0.34$ eV (above the valence band E_V). The concentrations of these defects proved to be dependent on Ga impurity concentration. In addition, the concentration of traps at $\approx E_V + 0.34$ eV grows in at $T \approx 200$ °C and peaks at $T \approx 300$ °C before the annealing stage over a temperature range of $T = 300$ – 400 °C.⁵ Generally speaking, this is in line with the EPR observation. Moreover, two important conclusions were drawn from an analysis of the available data. The role of carbon impurity atoms, especially for anneals at $T \geq 200$ °C, should be taken into consideration, whereas the formation of Ga-related defects is not strongly dependent on oxygen content.

Further photoluminescence studies (PL) on similar Ga-related defects in p -type Si provided support to the above conclusions.^{6–8} Three PL lines make their appearance in electron-irradiated Ga-doped silicon, designated as Ga1 (0.875), Ga2 (≈ 1.049), and Ga3 (0.926 eV). These luminescent Ga-related defects are formed and annealed out over a temperature range of interest, $100 \leq T \leq 400$ °C. Isotopic shifts $^{12}\text{C} \rightarrow ^{13}\text{C}$ revealed the incorporation of carbon atoms in the Ga1 and Ga3 centers. Unfortunately, the intensity of the Ga2 PL line turned out to be weak for detailed investigations. A comprehensive study of the singlet–triplet pair exciton emission has shown common electronic behavior of all three defects. A peculiar feature of the model discussed is based on the idea of pseudo-donors where a deeply bonded hole is recombined with a loosely binding electron emitting light. Actually, the binding energy of such holes is over a range of 100–280 meV, whereas for the loosely bound electrons, the binding energy is less than 20 meV, i.e., much less than the effective-mass value in the conduction band (above 30 meV). It has been shown that the orbital momentum of a deep hole bound in the Ga3 center is strongly quenched because of the large local strain associated with the defect.⁷ The simple model that meets the most experimental data collected so far is a pair of interstitial carbon and interstitial gallium atoms $[C_i Ga_i]$ sharing a substitutional lattice site along a $[001]$ axis. The optically active component in the pair is a single carbon atom. It is possible that the luminescent centers may incorporate more than one Ga atom, so in this way they are interrelated, displaying some slight changes in their atomic configurations.

Another well-known example of carbon-related quasi-chemical reactions in Ga-doped silicon is the formation of complexes with group-III acceptor impurities where two substitutional carbon and impurity atoms $[C_s Ga_s]$ are placed in the nearest lattice sites, so-called X-acceptors.⁹ The reason for the noticeable decrease in the ionization energy of the shallow acceptor states is the

increased electronegativity of the covalent Si–C bonds, replacing the ordinary Si–Si bonds.

A decade later, interest in gallium dopant has resurfaced in response to the search for an alternative to boron dopant in silicon-based solar cells. In accordance with this goal, intensive radiation experiments were carried out on silicon grown by the Czochralski technique [from here on p -Si(Cz):B and p -Si(Cz):Ga], i.e., crystals with high oxygen content.^{10,11} Six types of charge carrier traps were detected in 1 MeV electron-irradiated diode structures fabricated on p -Si(Cz):Ga. Among them, only new hole traps at $\approx E_V + 0.18$ eV in irradiated p -Si(Cz):Ga made their appearance during the annealing steps at around $T = 150$ °C, and then at higher annealing temperatures, their concentration raised by two orders-of-magnitude and peaked at $T = 325$ °C. Thereafter, these traps suddenly disappeared. Actually, among several other types of deep centers tentatively identified as containing oxygen and carbon atoms, only the centers under consideration could be attributed to Ga-related. It was noted that the annealing process of these defects is accompanied by a significant restoration of the free charge carrier concentration. This may indicate the annealing of deep radiation-induced donors. Furthermore, thermal emission of charge carriers was virtually independent of the bias. Based on the arguments presented, the authors concluded that these centers may be in an electrically inactive configuration at $T < 150$ °C. They become electrically active and thermally stable up to $T \approx 325$ °C only after annealing at $T \geq 150$ °C in a different configuration of donor-like complexes $[Ga_i O_i]$ or $[Ga_s Ga_i]$. Despite the great similarity of the formation and annealing of radiation defects in the investigations discussed with the similar behavior of radiation defects at $\approx E_V + 0.27$ eV in p -Si(FZ):Ga,⁵ the authors in Ref. 10 pointed to the absence of these centers in p -Si(Cz):Ga and explained the observed effect by suppression of the formation of defects at $\approx E_V + 0.27$ eV in the material due to high concentration of oxygen. The situation concerning charge carrier traps in irradiated p -Si:Ga seems to be more complicated because of many conflicting conclusions. To cite another example, the traps in electron-irradiated p -Si:Ga are reportedly at $\approx E_V + 0.16$ eV, $\approx E_V + 0.38$ eV, and $\approx E_V + 0.50$ eV (cf. Refs. 2, 5, and 10–12).

It is worth dwelling briefly upon the work where an original technique was used to study point defects such as self-interstitials and vacancies by hydrogen decoration. In this case, hydrogen atoms are prone to form complexes that consist of an intrinsic defect bound to two hydrogen atoms.¹³ By measuring the intensity of local vibrations of hydrogen atoms associated with vacancies and self-interstitials, this technique allows one to evaluate the efficiency of the Watkins replacement reaction for group-III impurities such as B, Al, Ga, and In. It was found that the efficiency of the impurity replacement reactions for B, Al, and Ga appears to be quite comparable in p -Si(FZ) irradiated with 3 MeV electrons at room temperature. Heavy indium impurity atoms exclusively exhibit significantly enhanced replacement reaction efficiency. Interestingly, the efficiency of the Watkins replacement reaction for boron turned out to be approximately proportional to the boron concentration at low fluences of the electron irradiation. By the way, it is appropriate to mention here that the same conclusion was made earlier when estimating the relative production rates of isolated vacancies and

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interstitial boron atoms in p -Si(FZ):B crystals subjected to 1.5 MeV electron irradiation at $T = 20.4$ K.¹⁴

To summarize this brief review, we can formulate some knotty issues associated with the current state of our knowledge of the electrical properties of point defects involving Ga impurity atoms in Si.

First, except for the well-known study¹ in which ion pairs $[Ga_s^- Ga_i^{2+}]$ have been reliably identified using EPR, the identification of Ga-related complexes in the studies cited above turns out to be indirect, simply relying on the use of this dopant in p -Si when modeling the formation and annealing processes of radiation defects. This ion pair is essentially the sole reliable example of establishing its atomic arrangements for the Ga-related complex in silicon, although its delayed formation only after an anneal of electron-irradiated p -Si:Ga at $T \approx 200$ °C has remained without a satisfactory explanation for a long time. This presents a striking contrast to boron-related complexes in p -Si for which a vast body of theoretical and experimental information is available nowadays, so their formation and annealing processes may be computer simulated in detail.

Second, the fundamental properties and thermal stability of secondary defects arising from quasi-chemical reactions of intrinsic point defects with Ga impurity atoms in irradiated p -Si:Ga at room temperature are scantily known. Because of the lack of reliable data, any attempt to formulate a quantitative model is still a challenging task.

Based on the above discussion, the aim of this study is to obtain *direct evidence* of the involvement of Ga impurity atoms in the formation and annealing of Ga-related defects in p -type Si irradiated with fast electrons and energetic protons at room temperature.

II. EXPERIMENTAL

The general technique of electrical measurements on Si samples is well established (see, for instance, Ref. 3 and references therein for details), and here, we will limit its description to the fundamental points.

In accordance with the stated goal, a strongly doped Si crystal was grown by the floating zone technique. The initial concentration of Ga impurity was about $N_{Ga} \approx 1 \times 10^{17} \text{ cm}^{-3}$. On the one hand, this has allowed us to strengthen the main formation processes of Ga-related defects in p -Si(FZ):Ga under irradiation. On the other hand, it helped us to lower the residual concentrations of oxygen and carbon impurities down to $N_{oxy} \leq 2 \times 10^{16}$ and $N_C \leq 6 \times 10^{15} \text{ cm}^{-3}$ to suppress any significant formation of oxygen- and carbon-related complexes with Ga impurity atoms.

Square-shaped samples of $7 \times 7 \text{ mm}^2$ and 0.3–0.4 mm in thickness were cut for electrical measurements in the geometry of van der Pauw technique. Hall effect and conductivity measurements were conducted over a wide temperature range of $T = 20$ –300 K. Experimental curves of the free charge carrier concentration and mobility, $p(1/T)$ and $\mu_h(T)$, respectively, were obtained in this way.

It is a matter of general experience^{15,16} that analysis of such $p(1/T)$ curves when the shallow acceptors are ionized (up to the saturation plateau) can be analyzed with the aid of charge balance

equations taking into account the statistics of holes in non-degenerate p -type Si. It makes possible the determination of the total concentration N_A of the substitutional Ga atoms, i.e., the shallow acceptor states, together with the total concentration N_D of all compensating donors being positively charged, both of which refer to conditions close to $T \approx 0$ K. Taking proper account of the complexity of the valence band of Si as well as the electronic structures of the shallow acceptor states, one can write the charge balance equation with separation of the variables related to all the concentrations on the left side and those tied to the parameters of the valence band and shallow acceptor states on the right side of the relevant equation,

$$\frac{p(p + N_D)}{N_A - N_D - p} = g_A^{-1} N_V \left(\frac{T}{T_0} \right)^{\frac{3}{2}} \exp \left(-\frac{E_A}{kT} \right). \quad (1)$$

Here, $N_V = N_V(T^{3/2})$ is the effective density-of-states in the valence band of Si, it is a power function of temperature, and k is Boltzmann's constant. E_A is the ionization energy of the ground state of the substitutional Ga atoms in the Si crystal lattice; it is deduced from the slope of experimental $p(1/T)$ curves at cryogenic temperatures; for the acceptor states of group-III impurities in Si, see Ref. 17. Together with the contributions of the valence sub-bands of light and heavy holes, the contribution of the split-off valence sub-band to the effective density-of-states of the valence band at elevated temperatures is taken into consideration as well. In accordance with this model, the degeneracy factors g_A of the ground state including the electron spin is taken into account too. Other details of calculations have been described in the literature.^{3,4} Under our experimental conditions, the Hall factor was assumed to be unity. Satisfactory agreement between the theoretical curves calculated by means of the charge balance equation given above and the experimental data bears witness to the validity of the model used. In doing so, it is possible to monitor the changes in N_A and N_D values in samples prior to and after irradiation, as well as in the course of subsequent anneals. The isochronal anneals were carried out in steps of $\Delta T = 40$ °C and $\Delta t = 10$ min, setting the reference temperature at $T = 20$ °C.

Samples were subjected to irradiation with a 3.5 MeV electron pulsed beam whose frequency was 200 cps. The duty cycle was 5 μ s. The electron flux density can be varied in the range of 1×10^{11} – $1 \times 10^{13} \text{ cm}^{-2} \text{ s}^{-1}$. For comparison purposes, several samples were also subjected to irradiation with a 15 MeV proton pulsed beam, whose frequency was 100 cps. The duty cycle was 2.5 ms. The samples were placed on a copper flange cooled by running water. Model radiation experiments with special samples, whose temperature was monitored by a thermocouple, showed that the temperature during proton irradiation did not exceed $T \approx 30$ °C. Furthermore, the average current was kept low, not exceeding 100 nA/cm^2 , to prevent heating of the samples during irradiation. Our estimates showed that the proton energy is high enough to pass through the irradiated samples of 0.3–0.4 mm thickness without noticeable effects of hydrogen-related passivation. The accuracy of the radiation dosage was about 15%.

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III. RESULTS AND DISCUSSION

A. Initial material

Electrical measurements of all initial samples allowed us to conclude that the strongly doped p -Si(FZ):Ga ($p \approx 1.1 \times 10^{17} \text{ cm}^{-3}$ at $T = 300 \text{ K}$) prior to irradiation was weakly compensated. The compensation ratio $K = N_d/N_A$ was very low, a few percent, so the hole mobility was around $\mu_h \approx 8 \times 10^3 \text{ cm}^2/\text{V s}$ at $T \approx 40 \text{ K}$.

B. Electron irradiation of p -Si:Ga

As a typical illustration, a profound effect produced by 3.5 MeV electron irradiation upon the hole concentration in p -Si(FZ):Ga is shown in Fig. 1 for one of the irradiated samples. In actual fact, the concentration of the shallow Ga acceptors was drastically dropped nearly by one-half as a result of electron irradiation, i.e., $\Delta N_A = N_A(\Phi) - N_A^0 = (4.6 \times 10^{16} - 1.1 \times 10^{17}) \text{ cm}^{-3} = -6.4 \times 10^{16} \text{ cm}^{-3}$ at a fluence of $\Phi = 6 \times 10^{16} \text{ cm}^{-2}$, where N_A^0 and $N_A(\Phi)$ are the total concentrations of the shallow acceptor states prior to and after electron irradiation, respectively. This provides *direct evidence* that intrinsic point defects in p -Si can interact with the substitutional Ga atoms at room temperature.

Along with a considerable decreasing concentration of the shallow acceptor states, the total concentration of the radiation-produced donors was found to be grown in, so one can assess the increasing compensation in the same way, $\Delta N_D = N_D(\Phi) \approx +3 \times 10^{16} \text{ cm}^{-3}$, taking into account that the initial concentration of compensating donors was negligibly small. It is worth noting

that the contribution of the donor states of divacancies under our irradiation conditions can be estimated as low as $2 \times 10^{15} \text{ cm}^{-3}$.¹⁸

Particular emphases are placed on a definite relationship between the N_A and N_D values. We found that $\Delta N_A(\Phi)$ appears to be twice as large as $\Delta N_D(\Phi)$ for all irradiated samples. We have every reason to believe that this is not accidental, since the loss of the shallow acceptor states is thought to consist of two main reactions between intrinsic point defects and substitutional Ga impurity atoms, which are deep concern to us.

Let us outline a general approach to the modern model of quasi-chemical defect reactions taking place in Ga-doped Si under electron irradiation. Primary point defects produced by irradiation are Frenkel pairs $[\text{Si}_i\text{-V}]$, i.e., self-interstitials Si_i bound to their parent vacancies V being somewhere nearby. It is well known that they are unstable and annihilated for the most part at room temperature. However, a fraction of them can escape from annihilation. Isolated intrinsic defects Si_i and V are very mobile, and they may be subjected to long-range migration in the crystal lattice. While migrating these defects, say, vacancies can be captured by substitutional Ga_s atoms, giving rise to the formation of vacancy-related complexes $[\text{Ga}_s\text{V}]$. In the case of self-interstitials Si_i trapped at substitutional Ga_s atoms, one may suggest the impurity replacement reaction proposed by Watkins,¹⁹ $\text{Si}_i + \text{Ga}_s \rightarrow [\text{Si}_i\text{Ga}_s] \rightarrow \text{Ga}_i + \text{Si}_s$ (a regular Si atom), with the resulting appearance of impurity interstitials; in this connection of Al-related defects, see Refs. 1 and 20. If other sinks of intrinsic point defects, e.g., oxygen and carbon atoms and so on, are inefficient, the Ga-related reactions may be assumed to proceed almost in a 1:1 correspondence. The relation experimentally established between the $\Delta N_A(\Phi)$ and $\Delta N_D(\Phi)$ values fits well with this reasonable model when taken into account that the loss of the shallow acceptor states comprises of two components associated with the formation of $[\text{Ga}_s\text{V}]$ pairs, on one hand, and impurity interstitials Ga_i , on the other hand.

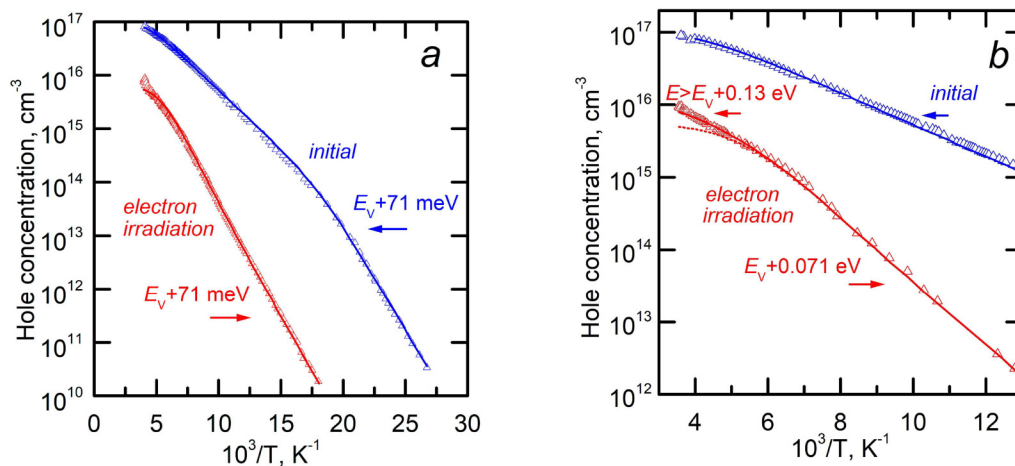


FIG. 1. Hole concentration vs reciprocal temperature for the strongly doped p -Si(FZ):Ga irradiated with 3.5 MeV electrons at room temperature (a). Fragments of the $n(10^3/T)$ curves are shown on the expanded temperature scale at $T > 78 \text{ K}$ (b). Points, experimental (in color online); curves, calculated. Points: prior to the irradiation (blue triangles); after the irradiation at a fluence of $\Phi = 6 \times 10^{16} \text{ electrons/cm}^2$ (red triangles). Effective ionization energies of shallow acceptors are given. The dotted line in (b) displays the saturation plateau of the shallow acceptors ionized, see text.

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There are strong grounds for believing that the electrical activity of the secondary radiation defects is different: $[Ga_sV]$ pairs are deep acceptors whose energy levels are thought to lie at $\approx E_V + 0.45$ eV,² whereas impurity interstitials Ga_i are deep donors.¹⁹ As has already pointed out, the presence of isolated impurity interstitials Ga_i^{2+} in electron-irradiated p -Si(CZ):Ga has never been identified by EPR,¹ contrary to the EPR detection of isolated interstitials Al_i^{2+} in electron-irradiated p -Si(FZ):Al.^{1,20} Among the energy levels detected by DLTS in electron-irradiated p -Si(CZ):Ga, the donor-like energy states at $\approx E_V + 0.18$ eV are deemed to be Ga-related.^{10,11} As is seen in Fig. 1(b), the hole concentration at $T \geq 250$ K still grows in the above saturation plateau due to the full ionization of the shallow acceptor states, meaning that there are other deep defects in the sample.

Isochronal anneals of the irradiated samples may tell one additional information on the thermal stability of the radiation-produced defects under consideration.

C. Isochronal annealing of defects in electron-irradiated p -Si:Ga

Some experimental and calculated curves of $p(1/T)$ obtained at some important annealing steps are depicted in Fig. 2. As can be seen from this figure, the first annealing stage of radiation defects was observed in a temperature interval of $T = 60$ – 100 °C. The second stage of partial recovery of the charge carrier concentration was found at around $T \approx 340$ °C, and the process of complete recovery occurred at higher temperatures of $T = 420$ – 560 °C.

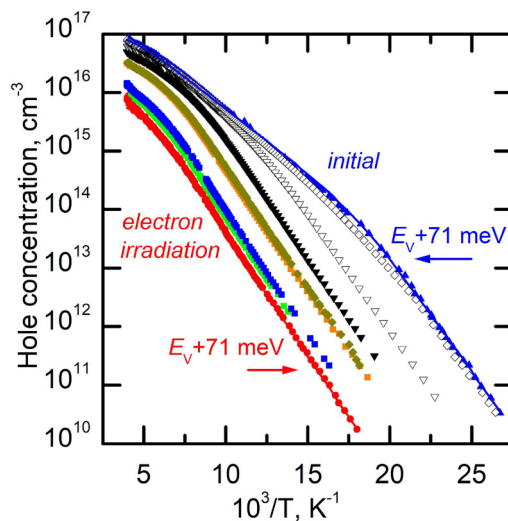


FIG. 2. Hole concentration vs reciprocal temperature for the strongly doped p -Si (FZ):Ga irradiated with 3.5 MeV electrons at room temperature and then subjected to isochronal annealing. Fluence Φ , electrons/cm²: 6×10^{16} . Curves, calculated. Points, experimental: initial (blue triangles up); irradiated (red circles); annealed at T : 60 °C (green squares); 140 °C (blue squares); 340 °C (orange squares); 380 °C (olive rhombi); 460 °C (black triangles down); 500 °C (white triangles down); 540 °C (white rhombi). Effective ionization energies of shallow acceptors are given.

An in-depth analysis based on the relevant equations of charge balance given above offers a means of furnishing information about the N_A and N_D variations in the course of annealing as well. This, in turn, can shed light upon the radiation defect behavior under heating (see Fig. 3).

As mentioned earlier, immediately after the electron irradiation, a slight decrease in the concentration of the shallow acceptor states occurred at relatively low temperatures, $T = 60$ – 100 °C. At the same time, the total concentration of compensating donors does not change at all. Significant changes in both concentrations become visible near $T \approx 300$ °C. After a short stabilization at $T = 340$ – 400 °C, the complete restoration of the shallow acceptor concentration to the initial value over a temperature range $T = 420$ – 560 °C is observed, together with the concentration of compensating donors returning to several 10^{14} cm⁻³.

Let us briefly discuss our preliminary identification of the main radiation defects involving Ga impurity atoms. First of all, this issue concerns pairs $[Ga_sV]$.

As we already mentioned in Sec. I, radiation defects annealed over a temperature range of $T = 90$ – 290 °C in Ref. 2 have been presumably attributed to such complexes in the course of a comparative study of p -Si(FZ):B, p -Si(FZ):Ga, and p -Si(FZ):Al subjected to 1.5 MeV electron irradiation at $T = 78$ and 300 K. It was found that at this annealing stage displaying the first order kinetics, the annealing activation energy is 0.42, 0.66, and 0.77 eV for B-, Ga-, and Al-related defects, respectively. It is worth recalling here that boron atom–vacancy pairs $[B_sV]$ are known to be unstable around room temperature due to some peculiarities in their atomic structure, and vacancy-related complexes $[Ga_sV]$ have not been detected by means of EPR yet. Therefore, the fact that EPR experiments on

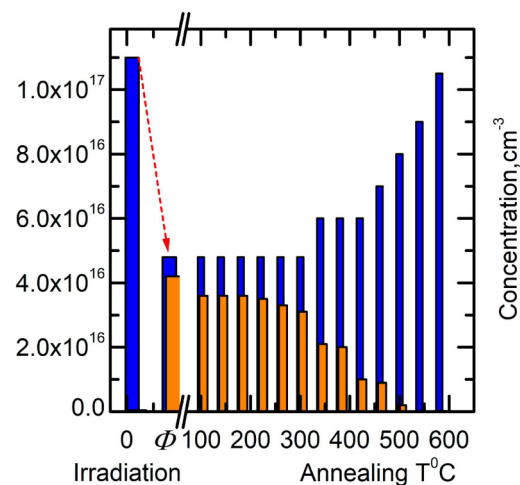


FIG. 3. Total concentrations of the shallow acceptor states (N_A) and deep donor states (N_D) for p -Si(FZ):Ga irradiated with 3.5 MeV electrons vs annealing temperature. Fluence Φ , electrons/cm²: 6×10^{16} . Concentrations of N_A and N_D are indicated by blue and orange bars, respectively. The change in the total concentration of the shallow acceptors after the electron irradiation is shown by the red arrow.

as-irradiated silicon p -Si(FZ):Al indicated the presence of stable vacancy-related pairs $[Al_iV]$ at $T \leq 300^\circ\text{C}$ may serve as a guide for us when the annealing activation energies given above are considered. Along with this, taking into account the predicted very small relaxation of the nearest neighboring atoms from the environment of the vacancy in the $[Ga_sV]$ complex being a mere 0.3% of their normal positions in the crystal lattice, the annealing stage at $T \approx 300\text{--}340^\circ\text{C}$ may be assigned to the disappearance of such pairs.²¹

However, the issue concerning a partial annealing of the vacancy-related defects at this stage remains an open question. Indeed, as can be seen from Fig. 3, only approximately one-half of the $[Ga_sV]$ pairs that disappeared at the annealing step are considered. One may assume that isolated and mobile vacancies released during the pair disintegration are trapped by other survived $[Ga_sV]$ complexes, thereby transforming them into more stable defects $[Ga_sVV]$, i.e., a divacancy anchored to a substitutional Ga atom. Such a defect complex is expected to be more stable than an isolated divacancy. Previously, a similar model supported by some theoretical arguments agrees with the annealing data satisfactorily when discussing the anneals of vacancy-related complexes in moderately doped p -Si(FZ)Si:B ($n \approx 5 \times 10^{15}\text{--}4 \times 10^{16}\text{ cm}^{-3}$) subjected to electron irradiation (see Ref. 3 and references therein). It should also be noted that at a later time, such a model of boron-related defects provides a satisfactory description of the restoration processes of electrical parameters during isochronal anneals of electron-irradiated silicon heavily doped with boron in a wide range of concentrations, from $p = 7.3 \times 10^{17}$ to $2.5 \times 10^{19}\text{ cm}^{-3}$.⁴

Let us move to a discussion of the high-temperature annealing stage of Ga-related defects. When considering the literature data on EPR,¹ we note that the upper temperature limit of the most stable radiation defects reliably identified as ion pairs $[Ga_s^- Ga_i^{2+}]$ is indicated as $T \approx 400^\circ\text{C}$, whereas in our case, the high-temperature annealing is spanned in an interval of $T \approx 420\text{--}560^\circ\text{C}$. We think that the annealing stage of these radiation-produced donors may be slightly shifted to higher temperatures due to several reasons. The Ga-doped p -Si crystal investigated in Refs. 1 and 14 was grown by the Czochralski technique. Together with this, the Ga impurity concentrations differ from one another by a factor of 4, about 3×10^{16} and $1.1 \times 10^{17}\text{ cm}^{-3}$ in Ref. 2 and in the present work, respectively. As a consequence, the damaging effect produced on the electron-irradiated Si:Ga samples in both experiments is expected to be much different, too. However, in this case, there are strong grounds for believing that we have to deal here with two kinds of Ga-related defects. Other than the ion pairs $[Ga_s^- Ga_i^{2+}]$ discussed above, a new kind of Ga-related defect not reported in the literature so far makes its appearance in the heavily irradiated samples. As can be seen from Fig. 3, these very stable defects being electrically neutral in the p -type material are present in high concentration, so their role is essential to the understanding of the observed changes in the electrical parameters.

Of even greater puzzling issue is the question that still remains without a reasonable explanation over a prolonged period. Unlike interstitial boron and aluminum ions that arise just after electron irradiation of p -Si and are studied making use of EPR, isolated Ga interstitials have never been observed in EPR experiments on irradiated silicon. The presence of ion pairs $[Ga_s^- Ga_i^{2+}]$ was detected

only after annealing of irradiated samples at $T \approx 180^\circ\text{C}$. The simple explanation that isolated Ga ions appear in the EPR-inactive charge state does not seem reasonable for a wide variety of p -Si:Ga samples studied and various irradiation conditions used.

The answer to this intriguing question was found in the peculiar behavior of the hole mobility $\mu_h(T)$ in the irradiated material.

Before moving on to the presentation of experimental curves of the temperature-dependent charge carrier mobility in silicon prior to and after irradiation, we consider it is necessary to make brief comments on the electrical measurements of electrical parameters using the van der Pauw technique. Square-shaped samples with small electrical contacts at the corners were investigated. Hall effect and resistivity measurements were carried out using four contacts in the pair arrangements. Experimental curves of the temperature-dependent charge carrier concentration $n, p(T)$ and conductivity $\sigma(T)$ as well as the charge carrier mobility $\mu(T)$ inevitably are slightly different because of some insignificant irregularities in the shape and thickness of each sample as well as electrical contact positions. More detailed information regarding such measurements and data processing can be found elsewhere (see, for instance, Ref. 22 and Figs. 3 and 4 therein). As to the charge carrier mobility $\mu(T)$, it is represented by two branches corresponding to the electrical measurements taken on two opposite sides of a tested sample. In normal cases, both branches are almost identical in appearance, but they are also slightly shifted relative to each other. The presence of substantial anisotropy of conductivity in irradiated samples results in the splitting of the branches $\mu(T)$ at cryogenic temperatures due to the scattering of charge carriers by the large elastic strains associated with radiation point defects.

As follows from Fig. 4, after the electron irradiation and subsequent annealing at $T \approx 180^\circ\text{C}$, any $\mu_h(T)$ curve at cryogenic temperatures was found to be split into two branches. Similar anisotropic effects of the charge carrier mobility $\mu_n(T)$ were observed in our study of radiation defects in the n -type floating zone Si doped with bismuth.²² The reason for this strange behavior of the mobility curves in irradiated n -Si(FZ):Bi is thought to be the unusual atomic configuration of the Bi atom–vacancy complex, in which the vacancy is split into a pair of half-vacancies with the Bi atom being at the saddle point.

Based on the above experimental information, there are reasons to assume that during the irradiation, mobile self-interstitials Si_i are trapped by substitutional Ga atoms that retain the captured intrinsic defects, thus forming pairs $[Si_iGa_i]$. This situation seems to be similar to the pairs of interstitial carbon and interstitial gallium atoms $[C_iGa_i]$ sharing a substitutional lattice site along a $[001]$ axis (see above in Sec. I). These defects are thought to be unobservable by EPR measurements. Such behavioral features distinguish Ga vs Al impurities in irradiated p -Si. In the latter case, the trapped self-interstitial replaces the substitutional Al atom in accordance with the Watkins model, $Si_i + Al_s \rightarrow Al_i + Si_s$ (a regular Si atom), leading to the formation of the isolated impurity interstitial in the tetrahedral interstitial site.^{1,19}

However, as mentioned earlier, in the present work, direct evidence has been obtained that impurity Ga atoms actively interact with intrinsic point defects just during electron irradiation at room temperature. Pairs $[Si_iGa_i]$ are stable up to $T \approx 180^\circ\text{C}$. After this point, the EPR experiments revealed the appearance of ion pairs

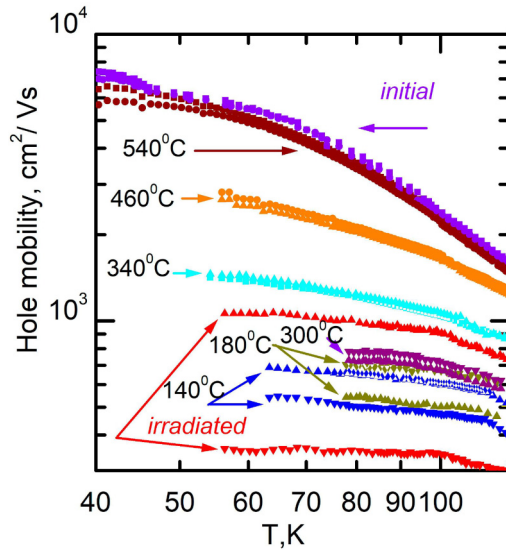


FIG. 4. Hole mobility vs temperature for p -Si(FZ):Ga irradiated with 3.5 MeV electrons at room temperature and then subjected to isochronal annealing. Fluence Φ , electrons/cm²: 6×10^{16} . Points, experimental: initial (violet squares and circles); irradiated (red triangles up and down); annealed at T : 140 °C (blue triangles up and down); 180 °C (dark yellow triangles up and down); 300 °C (purple triangles up and down); 340 °C (cyan triangles up and down); 460 °C (orange circles and triangles); 540 °C (wine squares and circles).

$[Ga_s^- Ga_i^{2+}]$ without the observation of isolated Ga interstitials. Therefore, it may be inferred that at $T \geq 200$ °C, the pairs $[Si_i Ga_i]$ under discussion are dissociated and the very mobile ions Ga_i^{2+} are captured by substitutional Ga atoms. As indicated by the anisotropic behavior of the charge carrier mobility $\mu_h(T)$, this process is completely finished at $T \approx 260$ °C.

It should also be added that, in our opinion, after an anneal at $T \approx 300$ °C, the total concentration of the compensating donors shown in Fig. 3 no longer includes the interstitial Ga-related complexes as such the pairs $[Si_i Ga_i]$ formed immediately upon the electron irradiation. It is assumed that in the irradiated sample annealed at $T \approx 300$ °C, the compensating donors are represented as ion pairs $[Ga_s^- Ga_i^{2+}]$ for the most part.

Nevertheless, the question arises as to what proportion of the ion pairs $[Ga_s^- Ga_i^{2+}]$ makes up in the total concentration of Ga-related defects annealed at $T \approx 400$ °C. An approximate estimate can be obtained from the diagram given in Ref. 19 for the fraction of ion pairs $[Al_s^- Al_i^{2+}]$ compared to the concentration of the isolated self-interstitials produced in p -Si(FZ):Al under irradiation with 1.5 MeV electrons at cryogenic temperatures. Again, it will be recalled that there is no experimental information concerning isolated interstitial Ga ions because of the known reasons (see above). Making use of the case of p -Si(FZ):Al as a reference, one can conclude that this fraction of the ion pairs $[Al_s^- Al_i^{2+}]$ is about one-third, which is thought to be true for irradiated p -Si doped with gallium, too. On the other hand, the conclusion implies that the majority of interstitial Ga atoms is involved in the formation of

very stable Ga-related complexes that are annealable at $T \geq 460$ °C. They are unobservable in EPR experiments but make their appearance in the present Hall effect measurements keeping watch on the restoration of the shallow acceptor states during high-temperature anneals. From Fig. 3, it can be concluded that such Ga-related complexes are electrically neutral in p -Si, for example, possessing deep acceptor states in the bandgap or being inert at all. Judging from the diagram of interstitial-related defects in irradiated Si,¹⁹ it is quite possible that interstitial Al impurity atoms in irradiated p -Si during high-temperature anneals may behave in a similar way (see also above).

It is noteworthy that the charge carrier mobility $\mu_h(T)$ in the p -Si(FZ):Ga immediately after irradiation and annealing at $T \leq 300$ °C was found to be not only very low at cryogenic temperatures $40 \leq T \leq 100$ K but it is getting almost temperature-independent as well (see Fig. 4). Such mobility characteristics for the available crystallographic orientation of the irradiated samples point to the very efficient hole scattering mechanism due to the large internal strain around the defects. Apparently, at higher annealing temperatures, the existing orientation of the studied samples seems to be nearly insensitive to the mobility anisotropy caused by the presence of ion pairs $[Ga_s^- Ga_i^{2+}]$ (see also below).

D. 15 MeV Proton irradiation of p -Si(FZ):Ga

Let us now discuss the experimental data obtained in studying the electrical parameters of p -Si(FZ):Ga irradiated with 15 MeV protons at room temperature. It is known that the general schema of main interactions between intrinsic point defects and dopant atoms in silicon under proton irradiations at relatively low energies $E \leq 20$ MeV does not qualitatively alter at moderate irradiation fluences, $\Phi \leq 5 \times 10^{14}$ protons/cm², but the rates of relevant reactions are substantially enhanced (see, for example, Refs. 3, 4, and 23).

As an example, Fig. 5 shows two curves of the hole concentration vs reciprocal temperature for one of the p -Si(FZ):Ga samples subjected to the proton irradiation.

Actually, the concentration of the shallow acceptors was drastically dropped as a result of proton irradiation, i.e., $\Delta N_A = N_A(\Phi) - N_A^0 = (2.0 \times 10^{16} - 1.1 \times 10^{17}) \text{ cm}^{-3} = -9 \times 10^{16} \text{ cm}^{-3}$ at a fluence of $\Phi = 5 \times 10^{14} \text{ cm}^{-2}$, where N_A^0 and $N_A(\Phi)$ are the total concentrations of the shallow acceptor states prior to and after the proton irradiation, respectively. Again, this provides convincing evidence that the substitutional Ga atoms interact with intrinsic point defects at room temperature. Worthy of mention are similar effects in the same material subjected to the electron irradiation too (see above). However, the removal rate of the shallow acceptor states appears to be larger by two orders-of-magnitude than that observed for the electron irradiation, $\eta_{Ga} \approx 180$ and 1.1 cm^{-1} , respectively.

The total concentration of radiation donors calculated by Eq. (1) was grown in, $\Delta N_D \approx N_D(\Phi) \approx +1.2 \times 10^{16} \text{ cm}^{-3}$ taking into account that the initial concentration of compensating donors was negligibly small. However, the compensation by the radiation-produced donors turned out to be very weak as compared to the observed loss of the shallow acceptors, $\Delta N_A = -9 \times 10^{16} \text{ cm}^{-3}$. The effect may somehow be associated with inhomogeneous distributions of primary defects in the proton tracks, contrary to the case

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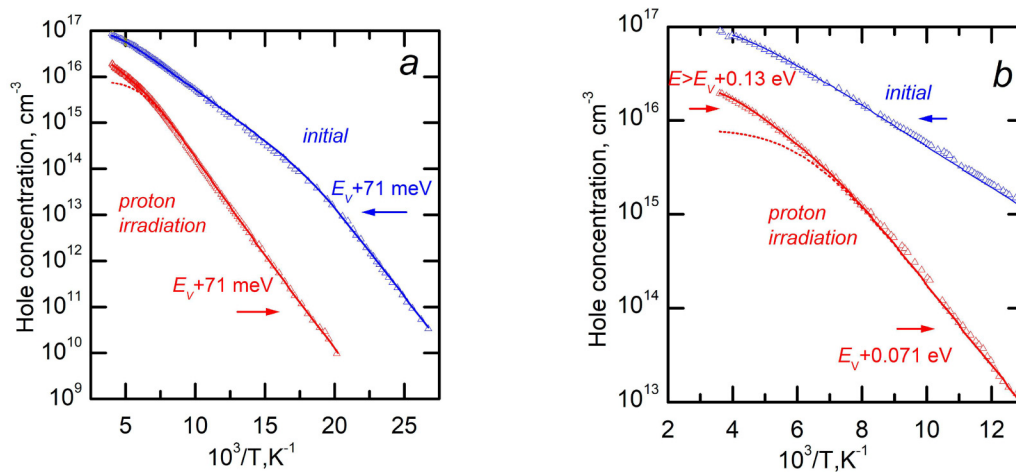


FIG. 5. Hole concentration vs reciprocal temperature for the strongly doped p -Si(FZ):Ga irradiated with 15 MeV protons at room temperature (a). Fragments of the $n(10^3/T)$ curves are shown on the expanded temperature scale at $T > 78$ K (b). Points, experimental (in color online); curves, calculated. Points: prior to the irradiation (blue triangles); after the irradiation at a fluence of $\Phi = 5 \times 10^{14}$ protons/cm² (red triangles). Effective ionization energies of shallow acceptors are given. The dotted line in (b) displays the saturation plateau of the shallow acceptors ionized, see text.

of electron irradiation. The presence of other deep radiation-related defects is also evident like in the case of the electron irradiation [cf. Figs. 1(b) and 5(b)].

E. Isochronal annealing of proton-irradiated p -Si:Ga

Some important annealing processes of the proton-irradiated sample are depicted in Fig. 6. As can be evident from this figure, the annealing of radiation defects below $T \leq 380$ °C was found to have an insignificant effect. A partial recovery of the charge carrier concentration was observed at around $T \approx 300$ – 340 °C, and the complete recovery of the charge carrier concentration occurred at higher temperatures from $T = 460$ to 560 °C.

Changes in the total concentration of acceptors and donors in one of the studied samples after proton irradiation and during isochronal anneals are shown in Fig. 7.

In general, the restoration of the shallow acceptors' concentration is akin to similar processes in the same material after electron irradiation (cf. Figs. 4 and 7). The annealing stage of the defects at $T \approx 300$ – 340 °C, which are thought to be pairs $[Ga_sV]$, as well as a much more pronounced high-temperature stage of Ga-related defects at $T \geq 420$ – 560 °C were found, too. It should be mentioned that such stable complexes formed under fast electron- and energetic proton irradiation of p -Si(FZ):Ga have never been reported in the literature (see also above).

However, now we are interested in an annealing temperature range of $T = 100$ – 200 °C, when the anisotropic behavior of the hole mobility $\mu_h(T)$ associated with $[Si_iGa_s]$ defects is usually observed (see Fig. 8). Just after the proton irradiation, the anisotropic effects lead to a split of the $\mu_h(T)$ curve into two branches whose positions do not noticeably change over an annealing interval indicated above. Such anisotropic effects turn out to be analogous to those discussed earlier for the electron-irradiated p -Si(FZ):Ga, as is seen

in Fig. 4. Thus, our conclusion on the formation of similar Ga-related complexes in the electron- and proton-irradiated p -Si (FZ):Ga thermally stable at $T \leq 200$ °C is confirmed. With increasing temperature, the impurity replacement reaction occurs,

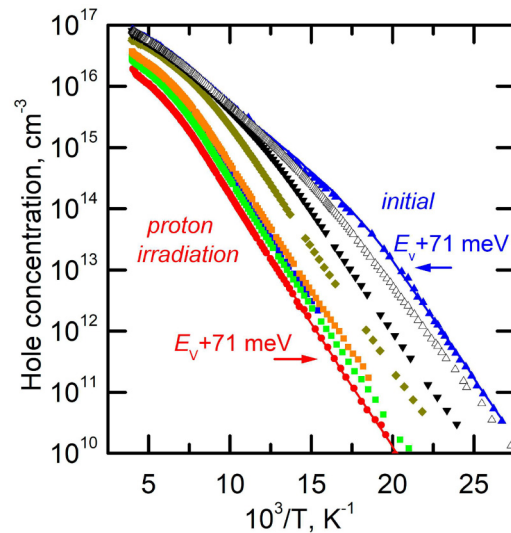


FIG. 6. Hole concentration vs reciprocal temperature for the strongly doped p -Si (FZ):Ga irradiated with 15 MeV protons at room temperature and then subjected to isochronal annealing. Fluence Φ , protons/cm²: 5×10^{14} . Curves, calculated. Points, experimental: initial (blue triangles up); irradiated (red circles); annealed at T : 180 °C (green squares); 260 °C (blue squares); 380 °C (orange squares); 460 °C (dark yellow rhombi); 500 °C (black triangles down); 540 °C (white triangles up). Effective ionization energies of shallow acceptors are given.

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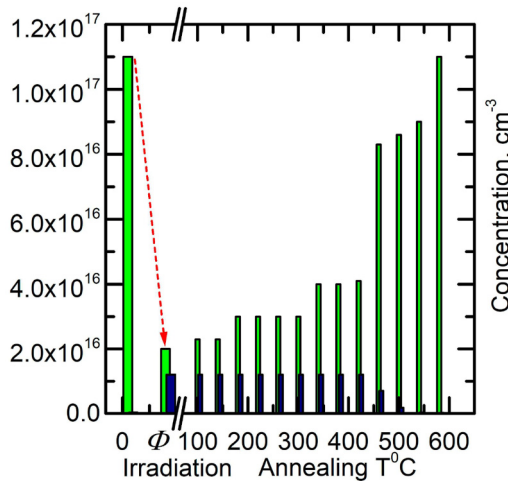


FIG. 7. Total concentrations of the shallow acceptor states (N_A) and deep donor states (N_D) for p -Si(FZ):Ga irradiated with 15 MeV protons vs annealing temperature. Fluence Φ , protons/cm²: 5×10^{14} . Concentrations of N_A and N_D are indicated by green and dark blue bars, respectively. The change in the total concentration of the shallow acceptors after the proton irradiation is shown by the red arrow.

resulting in the appearance of mobile ions Ga_i^{2+} , which, in turn, are captured by substitutional Ga_s^- atoms, as described above in Sec. III C. These ion pairs $[Ga_s^- Ga_i^{2+}]$ displaying donor activity are stable up to $T \approx 400^\circ\text{C}$, and thereafter, they disappear at higher annealing temperatures. It is also evident that a new kind of Ga-related complexes was proved to be unstable to annealing at $T \geq 460$ – 560°C , and thereafter, the electrical parameters of the proton-irradiated p -Si(FZ):Ga appear to be completely restored.

F. Anisotropy of conductivity of p -Si due to radiation-produced defects in p -Si

The important issue to be touched on is the conductivity behavior in irradiated and annealed samples. Two factors determining this parameter $\sigma(T)$ are dependent on the $p(T)$ and $\mu_h(T)$ functions. From the discussion above, one can now appreciate that anisotropic effects upon the conductivity are stemmed from the contribution of the last factor.

Before discussing the problem of conductivity anisotropy, the following remarks are necessary. To date, this issue has never been discussed in the literature, and no experimental data indicating the observation of conductivity anisotropy due to the presence of radiation defects in p -Si have been available, although some studies were carried out measuring the electrical parameters of irradiated Si samples using the Hall effect techniques. The focus of practical interest in the past was on the energy levels of radiation-produced defects in the bandgap as well as their thermal stability. Therefore, the crystalline orientation of samples was not taken into account in radiation experiments.

Moreover, much attention in that time was paid to radiation defects in n -type doped Si, in which vacancy-related complexes

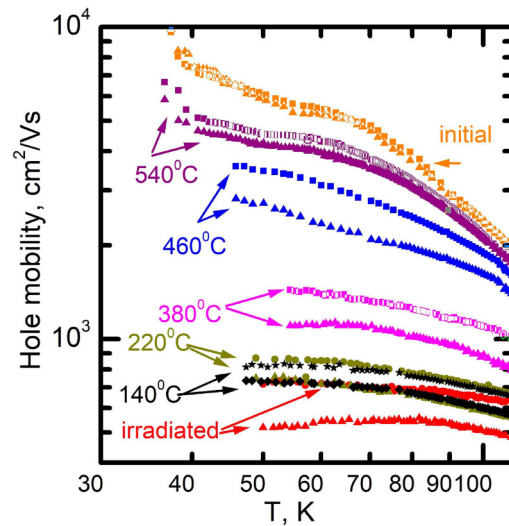


FIG. 8. Hole mobility vs temperature for p -Si(FZ):Ga irradiated with 15 MeV protons at room temperature and then subjected to isochronal annealing. Fluence Φ , protons/cm²: 5×10^{14} . Points, experimental: initial (olive squares and triangles up); irradiated (red circles and triangles up); annealed at T : 140°C (black stars and rhombi); 220°C (dark yellow triangles up and circles); 380°C (magenta squares and triangles up); 460°C (blue squares and triangles up); 540°C (purple squares and triangles up).

with Group V impurity atoms (P, As, and Sb) are electrically active. Nothing has been said as to conductivity anisotropy due to the formation of these complexes (often referred to as E -centers). Recently, it has been demonstrated that the anisotropic effect upon the $\sigma(T)$ curves due to substitutional phosphorus atom–vacancy pairs $[P_s V]$ turned out to be modest.²² Actually, the relaxation of the next-nearest neighboring atoms from the environment of the vacancy in the $[P_s V]$ complex is predicted by theoretical simulations to be around 0.1% of their normal positions in the crystal lattice. Nothing similar can be said about the pairs of $[As_s V]$ and $[Sb_s V]$ in silicon, which may display a larger atomic displacement around 2% and 8%, respectively.²¹ The three defect pairs mentioned retain a similar atomic structure with the vacancy as a whole defect and are deep acceptors in the n -type material. Until now, no one has reported a noticeable anisotropy of conductivity of irradiated Si doped with As and Sb. Self-interstitials and their complexes are believed not to play a significant role in changing the electrical parameters of irradiated n -Si if the concentrations of impurities such as oxygen and carbon are sufficiently low.²⁴

The situation changes radically in the bismuth–vacancy pair, when the displacement of the next-nearest neighboring atoms is approximately 11% and the presence of the $6f$ -electron shell will cause the vacancy to split into a pair of half-vacancies and the impurity atom to move to a saddle point. Together with this, the electrical activity of such vacancy-related complexes becomes neutral.²²

In striking contrast, interstitial impurities and their ion pairs in irradiated p -type Si, for example, $[B_s^- B_i^+]$ and so on, play an

important role.¹⁹ Such different quasi-chemical reactions of intrinsic point defects with group-V donors and group-III acceptors in Si subjected to electron and proton irradiation are thought to be associated with the dominant charge states of self-interstitials being neutral in the *n*-type material and positive in the *p*-type material.

Based on the analysis of the experimental data obtained in this study, it was concluded that the large local strains associated with radiation defects in the irradiated *p*-Si(FZ):Ga can be significant and highly inhomogeneous, causing the anisotropic behavior of the hole mobility $\mu_h(T)$ (see Figs. 4 and 8). These anisotropic effects manifest themselves at cryogenic temperatures of $T < 100$ K when the contribution of acoustic phonons in the charge carrier scattering is known to be very low. In contrast to vacancy-related complexes $[Ga_s V]$ produced only a slight relaxation of the next-nearest neighboring atoms,²¹ interstitial Ga-related defects can bring about the substantial local strains resulting in the anisotropic scattering of charge carriers at cryogenic temperatures.

Another example of this kind is the complexes involving boron impurity atoms in *p*-Si. Earlier, we studied *p*-Si(FZ):B under electron and proton irradiation.^{3,4} At that time, the well-known model concepts were used as a basis for interpreting the experimental data, including the $[B_s V]$ complex and the substitutional boron-interstitial boron ion pair $[B_s B_i^+]$. However, the problem of conductivity anisotropy was not addressed. For this reason, the issue of anisotropy in the present work was also tested on the *p*-Si(FZ):B subjected to 3.5 MeV electron irradiation. Figure 9 shows the hole mobility $\mu_h(T)$ in the *p*-Si(FZ):B after the irradiation and isochronal annealing in two characteristic regions of $T \approx 420$ °C and $T \geq 500$ °C.

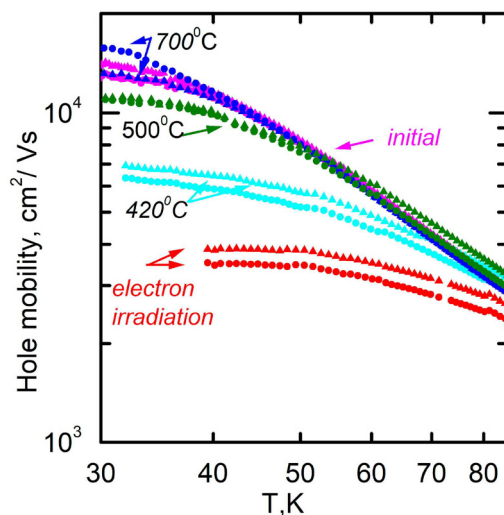


FIG. 9. Hole mobility vs temperature for *p*-Si(FZ):B irradiated with 3.5 MeV electrons at room temperature and then subjected to isochronal annealing. Fluence Φ , electrons/cm²: 4×10^{16} . Points, experimental: initial (magenta triangles up and circles); irradiated (red triangles up and circles); annealed at T : 420 °C (cyan triangles up and circles); 500 °C (olive triangles up and circles); 700 °C (blue triangles up and circles).

There are strong grounds for believing that the annealing of substitutional boron atom-divacancy complexes $[B_s VV]$ occurs at $T \approx 420$ °C. At higher annealing temperatures of $T \geq 500$ °C, ion pairs $[B_s^- B_i^+]$ are annealed out and the splitting of $\mu_h(T)$ curves disappears. It is worth noting that after the complete annealing of radiation defects at $T \approx 700$ °C, some anisotropy effects upon the hole mobility $\mu_h(T)$ observed at $T \leq 40$ K is thought to be due to the charge carrier scattering by the remaining ionized centers only.

Anisotropy effects turn out to be even more pronounced in *p*-Si(FZ):B after 8 MeV proton irradiation (see Fig. 10). With the available orientation of the irradiated sample, both the mobility branches were found to be split in a peculiar way; so one branch appears to be slightly below the initial hole mobility, whereas another one proved to be strongly changed.

In concluding this section, several remarks are necessary. The figures in this article, which for the first time demonstrate the presence of significant anisotropy of the charge carrier mobility in irradiated *p*-type silicon at cryogenic temperatures, refer to the non-oriented samples. When the charge carrier scattering in silicon irradiated at low radiation fluences is determined primarily by acoustic phonons over a temperature range of $T \geq 100$ K, the manifestation of anisotropy effects may be weak and mistakenly thought as Coulomb scattering by charged defects. In our case, the scattering mechanism of charge carriers due to the strain fields around radiation point defects is non-coulombic in nature. In order to correctly evaluate the characteristics of this scattering mechanism, future radiation experiments should be carried out on samples specially oriented along the main crystallographic axes.

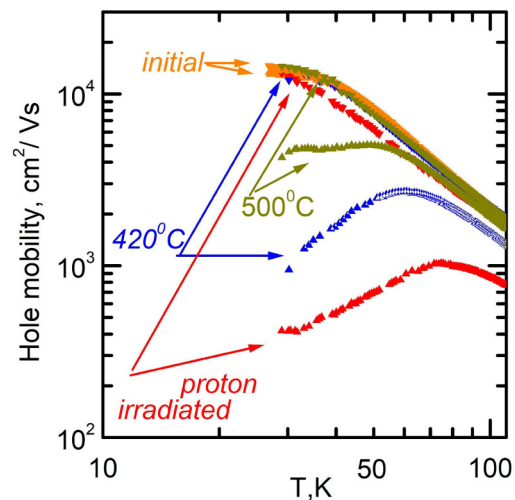


FIG. 10. Hole mobility vs temperature for *p*-Si(FZ):B irradiated with 8 MeV protons at room temperature and then subjected to isochronal annealing. Fluence Φ , protons/cm²: 1×10^{14} . Points, experimental: initial (orange triangles up and down); irradiated (red triangles up and down); annealed at T : 420 °C (blue triangles up and down); 500 °C (dark yellow triangles up and down).

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IV. CONCLUSIONS

In the present work, the main purpose was to obtain detailed information on the electrical properties of Ga-related defects that are formed in the p -Si(FZ):Ga under 3.5 MeV electron and 15 MeV proton irradiations. In contrast to some doubts concerning their formation at room temperature, the electrical measurements in a wide temperature range of $30 < T < 300$ K provided convincing evidence that Ga impurity atoms do interact with intrinsic point defects, bringing about the substantial loss of the shallow acceptor states as a result of the irradiation. Three kinds of Ga-related defects were found and preliminarily identified in the electron- and proton-irradiated samples. Among them, there is the vacancy-related complex $[Ga_sV]$, i.e., the substitutional Ga atom-vacancy pair. These pairs stable up to $T \approx 300$ °C are believed to be deep acceptors at $\approx E_V + 0.45$ eV. Together with this conclusion, there is also another defect, the substitutional-interstitial Ga ion pair $[Ga_s^- Ga_i^{2+}]$ earlier observed by EPR. These ion pairs are known to be stable up to $T \approx 400$ °C. Surprisingly, the presence of a new kind of interstitial Ga-related defects being electrically neutral was revealed in the irradiated p -Si(FZ):Ga. They appear to be very stable and can be annealed out at $T \geq 500$ °C. Moreover, these Ga-related defects in the heavily irradiated p -Si(FZ):Ga proved to be formed in substantial concentration.

The obtained data pointed to the fact that a striking effect of the anisotropic hole mobility $\mu_h(T)$ made their appearance after the irradiation of the p -Si(FZ):Ga as well the p -Si(FZ):B. In this case, the observed peculiarities of the hole mobility, in turn, are believed to be associated with the large inhomogeneous strain around the defects. With this in mind, we repeated electrical measurements of the hole mobility in the p -Si(FZ):B subjected to the electron and proton irradiations. The anisotropic behavior of the charge carrier mobility $\mu_h(T)$ in the irradiated p -Si(FZ):B at cryogenic temperatures was also confirmed. Thus, it can be concluded that this feature is characteristic of radiation-produced defects containing interstitial impurity ions. It is noteworthy that in an irradiated n -type material, similar anisotropic effects of the electron mobility $\mu_n(T)$ have been observed only in n -Si doped with oversized bismuth atoms.

Interestingly, the EPR detection of ion pairs $[Ga_s^- Ga_i^{2+}]$ only after an anneal of irradiated Ga-doped Si at $T \approx 180$ °C may be accounted for by first the formation of pairs containing two interstitials $[Si_iGa_i]$ stable at room temperature. It is appropriate to suggest that such a complex consisting of a self-interstitial tied to a Ga interstitial is unobservable by EPR. If the annealing temperature reaches $T \approx 200$ °C, the well-known impurity replacement reaction appears to be feasible, resulting in the appearance of very mobile interstitial Ga_i^{2+} ions, and thereafter in the formation of ion pairs $[Ga_s^- Ga_i^{2+}]$ enhanced by Coulomb attraction. The presence of pairs $[Si_iGa_i]$ during electron and proton irradiations is accompanied by the anisotropic behavior of the charge carrier mobility observed at lower temperatures.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Vadim Emtsev: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Nikolay Abrosimov:** Methodology (equal); Resources (equal); Writing – review & editing (equal). **Vitalii Kozlovski:** Investigation (equal); Methodology (equal); Resources (equal); Writing – review & editing (equal). **Stanislav Lastovskii:** Formal analysis (equal); Investigation (equal); Resources (equal); Writing – review & editing (equal). **Vadim Zhivul'ko:** Formal analysis (equal); Investigation (equal); Resources (equal); Software (equal); Writing – review & editing (equal). **Gagik Oganessian:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal). **Dmitrii Poloskin:** Data curation (equal); Software (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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